

Ziyan Liu

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<https://scholar.google.com/citations?user=JzleqTcAAAAJ>

Selected Research Experience

- **Watermarking Graph Neural Networks via Explanations for Ownership Protection** 07/2025 - 04/2026
Accepted by ICML 2026
 - As a co-first author, developed and validated a novel explanation-based GNN watermarking technique, demonstrating superior stealth and negligible impact on classification performance across diverse datasets.
 - Conducted a comprehensive comparative analysis against comparable explainer-based methodologies, distinctly isolating our framework's unique capacity.
 - Rigorously tested the robustness of the paradigm across a spectrum of challenging adversarial scenarios, including targeted fine-tuning attacks and adversarial perturbation.
- **Knowledge Graph based LLM Benchmark Construction Framework** 12/2025 - 02/2026
Accepted by KDD 2026
 - Built the first framework to automate LLM Benchmark question generation using any knowledge graph.
 - Research on impact of LLM performance from complexity of context and designed questions to evaluate the temporal reasoning capability.
 - Conducted prompt and context engineering to for LLM question generation and LLM-judge.
- **AdvQA: Query-Aware Adversarial Training for Knowledge Graph Reasoning** 03/2025 - 11/2025
Published in IEEE International Conference on Big Data
 - As the **sole first author**, proposed ADVQA, the first adversarially trained neural-symbolic framework for robust logical reasoning over Knowledge Graphs using Graph Neural Networks
 - Conducted extensive experiments on three benchmark datasets.
 - ADVQA achieves strong performance across standard FOL query types and significantly improves robustness under adversarial evaluation.

Work Experience

- **EvolutionIQ** 04/2025-10/2025
Senior Machine Learning Engineer
 - Built LLM-based insights extractor application from tens of millions of medical documents and insurance documentations
 - Spearheaded cross-functional initiatives to establish robust ground truth data pipelines, leveraging both human and LLM-based labeling, ultimately reducing false positives by 10% and reduce 30% manual time .
 - Developed context and prompt engineering strategy for LLMs to improve **contextual time-awareness** and medical diagnosis comprehension accuracy.
- **Opendoor** 10/2023-01/2025
Senior Machine Learning Engineer
Machine Learning Engineer 12/2020-09/2023
Machine Learning Modeling
 - Designed and productized of a new acquisition prediction model using the **Encoder Transformer** using **PyTorch** and improved **annual profit by 7.5 Million**
 - Designed model architecture and loss function to balance between model explainability and predictive performance
 - Developed a new underwriting sub-model using XGBoost and led the data quality improvement using active learning, data consolidation and weak supervision. Improved annual revenue by **150 Million**
 - Prototyped and later productized a multi-stage models pipeline to more efficiently utilize different groups of features in different stages of acquisition**MLOps**
 - Re-architected orchestration to be more flexible and saved **1.6 Million** model training cost annually
 - Designed and led the improvements of ML pipeline that saved **120 hours** from development monthly; tool stack includes AWS, AirFlow, Docker, Kubernetes.
 - Built a standardized model productization pipeline that consists of data quality monitoring, offline & online evaluation, canary test and A/B testing
 - Designed and led the implementation of CI/CD for independent release of code, model version and data schema .**New modeling Service Launch**
 - Led a cross functional team (Research, Pricing Agents, front-end) project to deprecate an old price prediction model used in Underwriting and increased annual profit by **30 Million** and saved annual cost by 500K
 - Translated ambiguous needs from partner team and promptly iterated on engineering goals

- Selected backend and API design considering customer experience, latency, extensibility and development effort
- Drove the decision-making on the launch plan considering online testing, metrics monitoring, and rollbacks
- Persuaded the PM to restrict testing within the underwriting phase by illustrating gains and risks via data analytics

- **Petuum, Inc. (Industrial ML, later spun out as oPRO.ai)**
Machine Learning Engineer

Sunnyvale, CA
04/2019-12/2020

Machine Learning Predictive Control

- Built deep learning models for **time series forecasting** used to automate production lines control
- Directly working with external clients such as Cemex and Phillips 66 and increased profit by **2.6 Million** for the clients from cost savings
- **Led meetings with domain experts and client-side engineering team** and translated customer needs into model design goals and metrics
- Achieved both accuracy and model interpretability by designing customized masking in **Pytorch**

Ranking System

- Built a **NLP based Information Retrieval System** for a professional network company (GuidePoint) with SOLR and PyTorch; saved manual efforts by 450 hours weekly across the company
- Discovered insights from user emails and selected proper models and methods of parsing and query expansion
- Utilized BM25f with creative query expansion to generate candidates and of **BERT** for ranking
- The system reduced hours of manual work to minutes with the same recommendation performance as humans

- **Canvs.ai**
Data Scientist

New York, NY
04/2017-04/2019

- Built NLP-based analytic products to provide insights from tweets and advertising campaign reviews
- Solved problems including text summarization, text-based sentiment classification, and named entity recognition

Education

- **Cornell University**
Master in Applied Statistics

January 2017
GPA 4.1/4.3

- Core Courses: Data Mining & Machine Learning, Natural Language Processing, Time Series Analysis, Linear Model with Matrices, Database Systems

- **Southwestern University of Finance and Economics**
Bachelor of Economics in Financial Engineering

June 2015
GPA 89/100

- Core Courses: C++, Mathematical Statistics, Stochastic Process, Financial Derivatives

Technical Skills

- **Programming Languages & Libraries:** Python (PyTorch, PySpark, Scikit Learn)
- **Containerization & orchestration Tools & Cloud Computing Platform:** AWS, Docker, Airflow, Kubernetes, Databricks
- **Search Engine and Databases:** ElasticSearch, SOLR, SQL